

PROBLEM STATEMENT #22 | Smart Cities, Transport & Logistics**Smart Parking Space Rental & License Validator****1. Problem Context & Overview**

A smart city parking system that handles real-time bay reservations, calculates progressive hourly rates, validates vehicle number plates via gate cameras, and flags unauthorized overstays.

Target Stakeholders / Actors: *Vehicle Owner, Parking Warden*

2. Sample Functional Requirement (FR) Guideline**Sample Requirement: FR-001 [Priority: High]**

Description: The system shall record vehicle entry time via license plate scan and compute total billing amount upon exit based on tiered hourly duration.

Sample Acceptance Criteria: Pass: Accurate fee calculation based on elapsed duration; Fail: Incorrect billing tier applied.

3. Sample Non-Functional Requirement (NFR) Guideline**Sample Requirement: NFR-001 [Type: Performance & Security]**

Description: The gate camera license plate lookup API shall verify parking reservation validity in under 300 ms.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]**• Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

• UML Use-Case Diagram:

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

• Use-Case Flow Specification:

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.